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* Multi-class Classification :-

Red - C_1
Blue - C_2
Green - C_3 } multi class (not binary)

$C=3$

① One vs Rest (0 vs 1)

② One vs One (0 vs 0)

① One vs Rest :-

→ so we converted multiclass model to binary class.

M1 :- Red vs (others) # so Red will be marked as "1"

or Red vs (Blue + Green) and rest will be -ve class or "0"

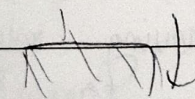
M2 :- Blue vs (others)

M3 :- Green vs (others)

x_1	x_2	y
1	8	Red
2	4.8	Blue
3.1	7.4	Blue
4.8	-4.2	Red
1.2	2.2	Green
2.1	-1.8	Blue

Test data $\Rightarrow 2.1, 8.4 \Rightarrow C=?$
which class?

"n" models for "n" classes.



These are binary classification models.

* The question we ask :- When we get new test data, which class do we assign it to?
∴ predict $\hat{y}$ given the test sample

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \Rightarrow \hat{y} = ?$$

sensitivity 1.0

10

logistic regression: Probability score $\rightarrow$ give threshold: then we get labels
 Decision trees: Labels.

* Score based models $\rightarrow$ which ones have we learnt?

$\rightarrow$ If our model gives us prob. scores as Red $\rightarrow$ 0.7, Blue $\rightarrow$ 0.2, Green $\rightarrow$ 0.9, we take Green as our label as it gives highest score and therefore highest confidence

$\rightarrow$ OVR relies on some kind of score to give out an output label similar to how important a particular node is in a classifier

Eg in SVM $\rightarrow$ margin is a score/metric

(highest score) $\arg \max_S (m \text{ score}) \rightarrow$ this is given as the label

* In classifiers we use some "metric" as a score.

(2) One vs One (O vs O)

M1: Red vs Blue $\rightarrow$ here we are removing rows with green and so on for each

M2: Red vs Green

$\therefore$ our dataset is getting shrunk this does not happen in OVR.

M3: Blue vs Green

* Handshake Problem
 If n classes, then how many models can I build? nC_2
 $\therefore$ higher models

- In pre-processing $\Rightarrow$ we will remove extra rows
- then we will train the model $\rightarrow$ build the model
- Now in our test phase, we want to predict $\hat{y}$. Therefore we pass through each model and take majority class (max voting)
- we also need to normalize

can also do $\rightarrow$ probability
 scoring
 $\rightarrow$ sum of prob. $\propto$ prob.
 for a label

* Feature selection :-

- minimal setup features that are good enough for a model.
 $\hookrightarrow$ we need minimal features to :
 - (i) reduce computational complexity

• No. of samples $\ll$ No. of features

$$S_1 \quad x_1, x_2, x_3 \dots \dots x_{100,000} \rightarrow y^{(1)}$$

$$S_2 \quad x_1, x_2, x_3 \dots \dots x_{100,000} \rightarrow y^{(2)}$$

$$S_3 \quad x_1, x_2, x_3 \dots \dots x_{100,000} \rightarrow y^{(3)}$$

* If more features, less samples $\rightarrow$ Overfitting / dependency

* Ideal no. of samples

$$ML: 10 \times \text{No. of features}$$

$$DL: 10000 \times \text{No. of features}$$

* Strategies :-

- ~~LASSO~~ / L1 Regularisation
- Forward Search
- Backward Search
- Recursive Feature Elimination (RFE)